SE-RGCL: Robust Graph Contrastive Learning against Adversarial Traffic Attacks in IoV

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APPENDIX

We compare SE-RGCL with a wide range of GNN baselines, from foundational architectures (GCN [1], GAT [2], GIN [3], GraphSAGE [4]) to more advanced models that handle deeper propagation (GCNII [5]), improve robustness (Grand [6]), and aggregate higher-order information (MixHop [7], $\mathrm{H_2GCN}$ [8]). The evaluation of all models is conducted on the NSL-KDD and UNSW-NB15 datasets without adding any data perturbation (*i.e.*, no adversarial attacks are implemented) to accurately measure their inherent performance under ideal conditions. As detailed in Table I and Table II, our proposed SE-RGCL exhibits a demonstrable advantage, consistently outperforming all other compared machine learning and deep learning models across key performance metrics and achieving superior overall performance.

TABLE I
Model Performance Evaluation on NSL-KDD Dataset.

Models	Acc	Precision	Recall	F1 score	AUC
GCN	0.7650	0.9210	0.6430	0.7570	0.9150
GAT	0.7440	0.9090	0.6700	0.7710	0.7900
GIN	0.8060	0.9530	0.6940	0.8030	0.8240
GraphSAGE	0.8000	0.9650	0.6730	0.7930	0.8200
GCNII	0.8550	0.9701	0.7850	0.8651	0.9335
Grand	0.9091	0.9750	0.8010	0.8821	0.9512
Mixhop	0.8750	0.9715	0.8552	0.9105	0.9400
$\mathrm{H_2GCN}$	0.9527	0.9726	0.8234	0.8500	0.9601
SE-RGCL	**0.9821**	**0.9824**	**0.9818**	**0.9820**	**0.9820**

TABLE II
Model Performance Evaluation on UNSW-NB15 Dataset.

Models	Acc	Precision	Recall	F1 score	AUC
GCN	0.8446	0.8404	0.8395	0.8399	0.8904
GAT	0.9131	0.9101	0.9002	0.9047	0.9104
GIN	0.9154	0.9091	0.9071	0.9081	0.9015
GraphSAGE	0.9193	0.9167	0.9072	0.9116	0.9132
GCNII	0.9225	0.9280	0.9135	0.9205	0.9195
Grand	0.9310	0.9355	0.9108	0.9270	0.9260
Mixhop	0.9216	0.9205	0.9255	0.9228	0.9295
$\mathrm{H_2GCN}$	0.9132	0.9085	0.9211	0.9013	0.8926
SE-RGCL	**0.9477**	**0.9538**	**0.9289**	**0.9398**	**0.9336**

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